



## Lesson 8.7 — Hypothesis Testing

*Big idea: A hypothesis test checks whether observed data is convincingly different from what we'd expect under a default (null) assumption. If the data would be too unlikely under the null, we reject it.*

### Quick review

**Null hypothesis ( $H_0$ ):** the “status quo” — usually no effect or no difference.

**Alternative hypothesis ( $H_a$ ):** what you're trying to find evidence for.

**Test statistic:** standardized distance from the null. For proportion:  $z = (p - p_0) / \sqrt{p_0(1 - p_0)/n}$ .

**p-value:** probability (under  $H_0$ ) of seeing a statistic as extreme as the one observed.

**Decision:** if p-value  $< \alpha$  (significance level, often 0.05), reject  $H_0$ ; otherwise fail to reject.

**Two error types:** Type I (reject true null, false alarm), Type II (fail to reject false null, missed detection).

### Worked example

**Worked example.** A coin is suspected of bias. You flip 100 times and get 60 heads. Test  $H_0: p = 0.5$  vs  $H_a: p \neq 0.5$  at  $\alpha = 0.05$ .

$z = (0.60 - 0.50) / \sqrt{(0.5 \cdot 0.5/100)} = 0.10/0.05 = 2.0$ . Two-tailed p-value =  $2 P(Z > 2) \approx 2(0.0228) \approx 0.046$ . Since  $0.046 < 0.05$ , reject  $H_0$ : evidence the coin is biased.

### Warm-ups

1. What is the null hypothesis if you're testing whether a new fertilizer changes yield?
2. Define the p-value in plain English.
3. If  $\alpha = 0.05$  and the p-value is 0.20, do you reject the null?

### Practice

1. A factory claims at most 5% defective. A sample of 200 has 16 defects. Test  $H_0: p = 0.05$  vs  $H_a: p > 0.05$  at  $\alpha = 0.05$ .
2. A test of  $H_0: \mu = 100$  vs  $H_a: \mu \neq 100$  has  $\bar{x} = 105$ ,  $s = 12$ ,  $n = 36$ . Find the test statistic.
3. Define Type I and Type II error in the context of a medical test for a disease ( $H_0$ : no disease).

### Challenge

1. A company claims its bottles average 12 oz. A random sample of 50 bottles has  $\bar{x} = 11.92$ ,  $s = 0.20$ . Test at  $\alpha = 0.05$  whether the bottles are underfilled. State  $H_0$ ,  $H_a$ , test statistic, p-value, conclusion.
2. Why is failing to reject the null NOT the same as proving the null true?



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## Answer key — Lesson 8.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

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### Warm-ups

1.  $H_0$ : the fertilizer has no effect on yield ( $\mu_{\text{new}} = \mu_{\text{old}}$ ).
2. Probability, assuming the null is true, of observing data at least as extreme as what we saw.
3. No — only reject when p-value  $< \alpha$ .

### Practice

1.  $p\blacksquare = 0.08$ .  $z = (0.08 - 0.05)/\sqrt{(0.05 \cdot 0.95/200)} = 0.03/\sqrt{0.0002375} \approx 0.03/0.01541 \approx 1.95$ . p-value  $\approx P(Z > 1.95) \approx 0.0256 < 0.05 \Rightarrow$  reject. Evidence that defect rate exceeds 5%.
2.  $z = (105 - 100)/(12/\sqrt{36}) = 5/2 = 2.5$ .
3. Type I: claim someone has the disease when they don't (false positive). Type II: claim someone is healthy when they actually have the disease (false negative).

### Challenge

1.  $H_0: \mu = 12$ .  $H_a: \mu < 12$  (one-sided).  $z = (11.92 - 12)/(0.20/\sqrt{50}) = -0.08/(0.0283) \approx -2.83$ . p-value  $= P(Z < -2.83) \approx 0.0023 < 0.05$ . Reject. Evidence that bottles are underfilled.
2. Failing to reject means the data are consistent with  $H_0$ , but consistent doesn't mean uniquely supported. The data could also be consistent with many alternatives close to  $H_0$ . Absence of evidence isn't evidence of absence.